

SOC SECURITY MONITORING USING SPLUNK ENTERPRISE

ABSTRACT

Security monitoring has become an essential component of modern cybersecurity due to the increasing sophistication of cyber threats. Organizations require centralized monitoring systems capable of collecting, analyzing, and correlating security events generated across multiple endpoints.

This project presents the implementation of a Security Operations Center (SOC) laboratory using Splunk Enterprise as the Security Information and Event Management (SIEM) platform. Microsoft Sysmon was integrated with Windows Event Logs to provide enhanced endpoint visibility, while Splunk Universal Forwarder was configured to securely transmit log data to the Splunk Enterprise server.

Custom Search Processing Language (SPL) queries were developed to detect PowerShell execution, failed login attempts, DNS queries, process creation, network connections, and suspicious process execution. Automated alerts were configured for each detection rule, and interactive dashboards were designed to provide centralized monitoring of security events. Furthermore, all detection rules were mapped to the MITRE ATT&CK Framework to align the monitoring environment with industry-standard threat classification.

The implementation successfully demonstrated centralized log collection, endpoint monitoring, threat detection, alert generation, and security visualization. The developed SOC laboratory provides practical experience in SIEM deployment, incident investigation, and threat monitoring while closely simulating real-world Security Operations Center activities.

Keywords: Security Operations Center (SOC), SIEM, Splunk Enterprise, Sysmon, Windows Event Logs, Threat Detection, MITRE ATT&CK, Cybersecurity, Incident Response.

CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's digital era, organizations increasingly rely on interconnected computer systems, cloud platforms, and internet-based services to conduct their daily operations. While technological advancements have significantly improved efficiency and productivity, they have also introduced new cybersecurity challenges. Cyber threats such as malware infections, ransomware attacks, phishing campaigns, insider threats, unauthorized access, privilege escalation, and advanced persistent threats (APTs) have become more frequent

and sophisticated. These attacks can compromise sensitive information, disrupt business operations, and cause significant financial and reputational losses.

Traditional security solutions such as antivirus software, firewalls, and intrusion prevention systems provide a basic level of protection but are often insufficient to detect modern cyber threats. Attackers continuously develop new techniques to evade conventional security controls, making it essential for organizations to implement continuous monitoring and proactive threat detection mechanisms. Consequently, Security Operations Centers (SOCs) have become a fundamental component of modern cybersecurity strategies.

A Security Operations Center (SOC) is a centralized facility responsible for continuously monitoring, analyzing, detecting, investigating, and responding to cybersecurity incidents across an organization's IT infrastructure. A SOC combines people, processes, and technology to identify malicious activities before they cause significant damage. Security analysts working within a SOC collect logs from multiple systems, correlate security events, identify indicators of compromise, investigate suspicious activities, and initiate incident response procedures when required.

Security Information and Event Management (SIEM) solutions play a critical role in SOC operations. SIEM platforms collect logs from multiple sources, normalize event data, correlate related events, generate alerts, and provide real-time visualization of security incidents. Among various SIEM solutions available today, Splunk Enterprise has emerged as one of the most widely adopted platforms because of its powerful search capabilities, real-time analytics, customizable dashboards, and scalability.

This project focuses on designing and implementing a Security Operations Center using Splunk Enterprise as the SIEM platform. Windows Security Event Logs and Sysmon telemetry are collected from a Windows endpoint through Splunk Universal Forwarder and centrally analyzed within Splunk Enterprise. Custom detection rules have been developed to detect various attacker activities including brute-force login attempts, PowerShell execution, command prompt execution, account discovery, process discovery, network reconnaissance, DNS activity, and registry modifications.

To enhance the effectiveness of threat detection, all implemented detection rules have been mapped to the MITRE ATT&CK Framework. The framework provides standardized adversarial tactics and techniques that enable security analysts to better understand attacker behavior and improve detection capabilities.

The project also includes the simulation of multiple cyberattack scenarios using a Kali Linux virtual machine. These simulations validate the implemented detection rules and demonstrate how a SOC can effectively detect suspicious activities through centralized log monitoring, automated alerts, and interactive dashboards.

Overall, this project demonstrates the practical implementation of a Security Operations Center capable of collecting endpoint telemetry, performing security event correlation, generating alerts, and providing real-time visibility into endpoint activities within a controlled laboratory environment.

1.2 Background of the Study

The increasing dependence on digital technologies has transformed the way organizations operate. Business operations, financial transactions, healthcare services, educational institutions, and government agencies rely heavily on interconnected information systems. While digital transformation has introduced numerous benefits, it has also expanded the attack surface available to cybercriminals.

Cyberattacks have evolved from simple viruses and worms to sophisticated campaigns involving ransomware, fileless malware, credential theft, lateral movement, and privilege escalation. Many of these attacks remain undetected for extended periods because organizations lack effective monitoring mechanisms capable of identifying malicious behavior in real time.

To address these challenges, organizations have established Security Operations Centers (SOCs) that continuously monitor enterprise environments. SOC analysts utilize Security Information and Event Management (SIEM) platforms to collect and analyze logs generated by operating systems, applications, firewalls, intrusion detection systems, endpoint detection solutions, and network devices.

Microsoft Windows operating systems generate large volumes of security logs that provide valuable information about authentication attempts, user activities, system events, and security-related operations. However, native Windows Security Logs alone may not provide sufficient visibility into advanced attacker techniques.

Sysmon (System Monitor), developed by Microsoft Sysinternals, extends Windows logging capabilities by generating detailed telemetry related to process creation, network connections, registry modifications, file creation, driver loading, DNS queries, and numerous other system activities. When combined with Windows Event Logs, Sysmon significantly improves endpoint visibility and supports effective threat detection.

Splunk Enterprise serves as the centralized platform responsible for collecting, indexing, searching, correlating, and visualizing security events. By integrating Windows Security Logs, Sysmon telemetry, and custom detection rules, organizations can rapidly identify suspicious behavior and respond to security incidents more effectively.

This project aims to demonstrate how these technologies can be integrated to build a practical SOC environment capable of detecting multiple stages of cyberattacks in real time.

1.3 Problem Statement

Modern organizations generate enormous volumes of security logs every day. Manual analysis of these logs is impractical and often leads to delayed detection of cyber threats. Traditional security mechanisms primarily focus on prevention but frequently fail to identify sophisticated attacks that bypass perimeter defenses.

Many organizations also struggle with fragmented security monitoring due to the absence of centralized log collection and real-time analytics. Without an effective SIEM platform, suspicious activities such as unauthorized login attempts, malicious PowerShell execution, privilege escalation, account enumeration, registry modifications, and network reconnaissance may remain unnoticed until significant damage has occurred.

Therefore, there is a need to design and implement a centralized Security Operations Center capable of collecting endpoint telemetry, correlating security events, generating automated alerts, and providing real-time visibility into malicious activities using industry-standard technologies.

1.4 Objectives of the Project

The primary objective of this project is to design and implement a Security Operations Center capable of detecting and monitoring cybersecurity threats using Splunk Enterprise.

The specific objectives are:

- To install and configure Splunk Enterprise as the centralized SIEM platform.
- To configure Splunk Universal Forwarder for secure log forwarding.
- To install and configure Microsoft Sysmon for enhanced endpoint telemetry.
- To collect Windows Security Event Logs and Sysmon logs.
- To design an interactive SOC dashboard for security monitoring.
- To develop custom detection rules for identifying suspicious activities.
- To configure automated alerts for high-priority security events.
- To map detection rules to the MITRE ATT&CK Framework.
- To simulate cyberattacks using Kali Linux.
- To evaluate the effectiveness of the implemented SOC through attack simulations.

1.5 Scope of the Project

The scope of this project is limited to the implementation of a laboratory-based Security Operations Center for cybersecurity monitoring and threat detection.

The project includes:

- Deployment of Splunk Enterprise.
- Deployment of Splunk Universal Forwarder.
- Configuration of Windows Event Log collection.
- Installation of Microsoft Sysmon.

- Development of custom detection rules.
- Real-time dashboard visualization.
- Alert generation.
- MITRE ATT&CK mapping.
- Controlled cyberattack simulations.

The project does not include enterprise-scale deployment, cloud-based SIEM implementation, SOAR integration, or malware reverse engineering.

1.6 Project Methodology

The project follows a systematic implementation methodology consisting of the following phases:

1. Requirement Analysis
2. Environment Setup
3. Splunk Enterprise Installation
4. Universal Forwarder Configuration
5. Sysmon Deployment
6. Log Collection and Validation
7. Dashboard Development
8. Detection Rule Development
9. Alert Configuration
10. Attack Simulation
11. Result Analysis
12. Documentation

This structured methodology ensures accurate implementation, testing, and evaluation of the SOC environment.

1.7 Organization of the Report

This report is organized into ten chapters to provide a structured explanation of the project.

- Chapter 1 introduces the project, objectives, scope, methodology, and background.

- Chapter 2 presents the literature review covering SOCs, SIEM, Splunk Enterprise, Sysmon, and the MITRE ATT&CK Framework.
- Chapter 3 describes the overall system architecture and design.
- Chapter 4 explains the implementation process, including Splunk installation, Universal Forwarder configuration, Sysmon deployment, and log collection.
- Chapter 5 discusses the design and implementation of the SOC dashboard.
- Chapter 6 presents the custom detection rules and alert configurations.
- Chapter 7 explains the MITRE ATT&CK mapping and attack simulations.
- Chapter 8 presents the results and discussion.
- Chapter 9 provides the conclusion and recommendations for future enhancements.
- Chapter 10 includes references and appendices.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Cybersecurity has become one of the most critical areas of information technology due to the increasing number and sophistication of cyber threats. Organizations across the world rely heavily on digital infrastructure to support business operations, making them attractive targets for cybercriminals. Traditional security mechanisms such as antivirus software and firewalls provide the first line of defense but are no longer sufficient to detect modern attacks that employ advanced evasion techniques. As a result, organizations have shifted towards continuous monitoring, centralized log management, and proactive threat detection through Security Operations Centers (SOCs).

A SOC enables organizations to monitor their infrastructure continuously, detect malicious activities, investigate security incidents, and respond before significant damage occurs. The effectiveness of a SOC largely depends on the implementation of Security Information and Event Management (SIEM) platforms, endpoint monitoring tools, and structured threat intelligence frameworks. This chapter reviews the literature related to SOCs, SIEM solutions, Splunk Enterprise, Microsoft Sysmon, Windows Event Logging, and the MITRE ATT&CK Framework.

2.2 Security Operations Center (SOC)

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, investigating, and responding to cybersecurity incidents. It combines people, technology, and well-defined processes to provide continuous visibility into an

organization's security posture. SOC analysts monitor alerts, analyze logs, investigate suspicious behavior, and coordinate incident response activities.

Modern SOC's operate around the clock and collect security data from multiple sources, including operating systems, firewalls, intrusion detection systems, endpoint protection platforms, web servers, cloud services, and network devices. This centralized approach allows analysts to correlate events from different systems and identify attack patterns that might otherwise remain unnoticed.

The primary responsibilities of a SOC include:

- Continuous security monitoring
- Threat detection
- Log analysis
- Incident investigation
- Threat hunting
- Incident response
- Security reporting
- Compliance monitoring

The increasing volume of security events generated by enterprise environments makes manual monitoring impossible. Therefore, automated log collection, event correlation, and alert generation have become essential components of modern SOC operations.

2.3 Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) is a cybersecurity solution that collects, stores, analyzes, and correlates security events generated by various systems and applications. SIEM platforms enable organizations to centralize security monitoring while providing advanced search capabilities, dashboards, automated alerts, and forensic investigation tools.

The primary functions of a SIEM include:

- Centralized log collection
- Log normalization
- Event correlation
- Security analytics
- Dashboard visualization
- Automated alert generation

- Incident investigation
- Compliance reporting

Modern SIEM solutions improve threat detection by correlating events from multiple sources instead of analyzing individual logs independently. This capability significantly reduces the time required to identify suspicious activities.

Several commercial and open-source SIEM platforms are available, including Splunk Enterprise, IBM QRadar, Microsoft Sentinel, Elastic Security, and Wazuh. Among these, Splunk Enterprise has gained widespread adoption because of its scalability, powerful search language, flexible dashboards, and extensive integration capabilities.

2.4 Splunk Enterprise

Splunk Enterprise is one of the most widely used SIEM platforms for collecting, indexing, searching, analyzing, and visualizing machine-generated data. It supports real-time monitoring by collecting logs from endpoints, servers, applications, databases, and network devices.

One of the major strengths of Splunk is its Search Processing Language (SPL), which enables analysts to perform complex searches, extract fields, correlate events, generate statistics, and create dashboards. SPL provides flexibility in developing custom detection rules that can identify suspicious activities across different environments.

Key features of Splunk Enterprise include:

- High-speed log indexing
- Real-time search capabilities
- Interactive dashboards
- Automated alerting
- Data visualization
- Event correlation
- Threat detection
- Scalable architecture

In this project, Splunk Enterprise serves as the centralized SIEM responsible for collecting Windows Security Event Logs and Sysmon telemetry through Splunk Universal Forwarder. Custom SPL queries were developed to detect suspicious activities and generate alerts.

2.5 Microsoft Sysmon

Microsoft Sysmon (System Monitor) is a Windows system service developed by Microsoft Sysinternals that provides advanced endpoint telemetry. Unlike traditional Windows Event Logs, Sysmon records detailed information about system activities that are highly valuable during incident investigations and threat detection.

Sysmon generates detailed events related to:

- Process creation
- Network connections
- Driver loading
- Image loading
- Registry modifications
- File creation
- File deletion
- DNS queries
- Process access

Each event generated by Sysmon contains extensive contextual information such as process identifiers, parent processes, command-line arguments, user accounts, hashes, timestamps, and network details. These events enable SOC analysts to detect sophisticated attacker techniques that might otherwise remain hidden.

In this project, Sysmon significantly enhanced endpoint visibility and served as the primary source of security telemetry used for developing custom detection rules.

2.6 Windows Security Event Logs

Windows operating systems generate Security Event Logs that record authentication events, account management activities, policy changes, system events, and other security-related operations. These logs provide valuable information regarding user authentication and system access.

Important Windows Security Event IDs include:

- Event ID 4624 – Successful Login
- Event ID 4625 – Failed Login
- Event ID 4634 – Logoff
- Event ID 4648 – Explicit Credential Logon
- Event ID 4672 – Special Privileges Assigned
- Event ID 4688 – Process Creation

Windows Event Logs are widely used by SOC analysts to identify brute-force attacks, unauthorized access attempts, privilege escalation, and suspicious user behavior.

The combination of Windows Security Event Logs and Sysmon telemetry provides comprehensive endpoint monitoring capabilities.

2.7 Splunk Universal Forwarder

Splunk Universal Forwarder is a lightweight agent responsible for securely forwarding logs from endpoints to a centralized Splunk Enterprise server. It consumes minimal system resources while ensuring reliable and secure log transmission.

The Universal Forwarder supports the collection of:

- Windows Event Logs
- Sysmon Logs
- Application Logs
- Custom Log Files
- Security Logs

In this project, the Universal Forwarder was installed on the Windows endpoint to forward Windows Security Event Logs and Sysmon Operational Logs to Splunk Enterprise for centralized analysis.

2.8 MITRE ATT&CK Framework

The MITRE ATT&CK Framework is a globally recognized knowledge base that documents adversary tactics, techniques, and procedures (TTPs) observed during real-world cyberattacks. Instead of focusing on malware families, MITRE ATT&CK emphasizes attacker behavior throughout the attack lifecycle.

The framework categorizes attacker activities into tactical stages such as:

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery
- Lateral Movement

- Collection
- Command and Control
- Exfiltration
- Impact

Security analysts use MITRE ATT&CK to improve detection coverage, identify security gaps, prioritize defensive measures, and standardize threat reporting.

In this project, each custom detection rule was mapped to the appropriate MITRE ATT&CK technique to provide context regarding attacker behavior and improve analytical capabilities.

2.9 Related Work

Several researchers have demonstrated the effectiveness of SIEM platforms in improving organizational security monitoring. Studies show that centralized log collection significantly reduces incident response time while improving threat visibility.

Research on Splunk Enterprise highlights its capability to process large volumes of machine data, perform real-time analytics, and generate customized dashboards for security monitoring. Similarly, Microsoft Sysmon has been widely recognized as an effective endpoint monitoring solution that provides detailed telemetry for detecting advanced attacker techniques.

Previous studies have also emphasized the importance of combining Windows Event Logs, Sysmon telemetry, SIEM platforms, and the MITRE ATT&CK Framework to develop comprehensive threat detection capabilities. These approaches enable SOC analysts to identify malicious activities with greater accuracy while reducing false positives.

The present project builds upon these concepts by integrating Splunk Enterprise, Sysmon, Windows Event Logs, and MITRE ATT&CK into a practical SOC laboratory capable of detecting multiple attack scenarios through custom SPL detection rules and automated alerts.

2.10 Chapter Summary

This chapter reviewed the fundamental concepts related to Security Operations Centers, SIEM technology, Splunk Enterprise, Microsoft Sysmon, Windows Event Logs, Splunk Universal Forwarder, and the MITRE ATT&CK Framework. It also discussed previous research demonstrating the effectiveness of centralized log management and endpoint monitoring for cybersecurity operations.

The reviewed literature establishes the theoretical foundation required for implementing the SOC environment developed in this project. The next chapter describes the overall system architecture and design of the proposed Security Operations Center.

CHAPTER 3

SYSTEM DESIGN AND ARCHITECTURE

3.1 Introduction

The successful implementation of a Security Operations Center (SOC) requires a well-structured architecture that enables centralized log collection, security monitoring, threat detection, and incident analysis. This chapter presents the architecture and design of the SOC Security Monitoring system developed in this project. The implemented environment integrates Splunk Enterprise, Splunk Universal Forwarder, Microsoft Sysmon, and Windows Event Logs to provide centralized visibility into endpoint activities.

The architecture was designed to simulate a real-world SOC environment where endpoint telemetry is continuously collected, indexed, analyzed, and visualized through interactive dashboards. Security events generated on the Windows endpoint are forwarded to Splunk Enterprise, where custom Search Processing Language (SPL) queries analyze the data and generate alerts whenever suspicious activities are detected.

3.2 System Architecture

The SOC environment consists of three major components:

1. Windows 10 Endpoint
2. Splunk Enterprise Server
3. Kali Linux Virtual Machine

The Windows endpoint acts as the monitored system where user activities and security events are generated. Microsoft Sysmon is installed to enhance Windows logging capabilities by recording detailed endpoint telemetry such as process creation, network connections, DNS queries, registry modifications, and file activities.

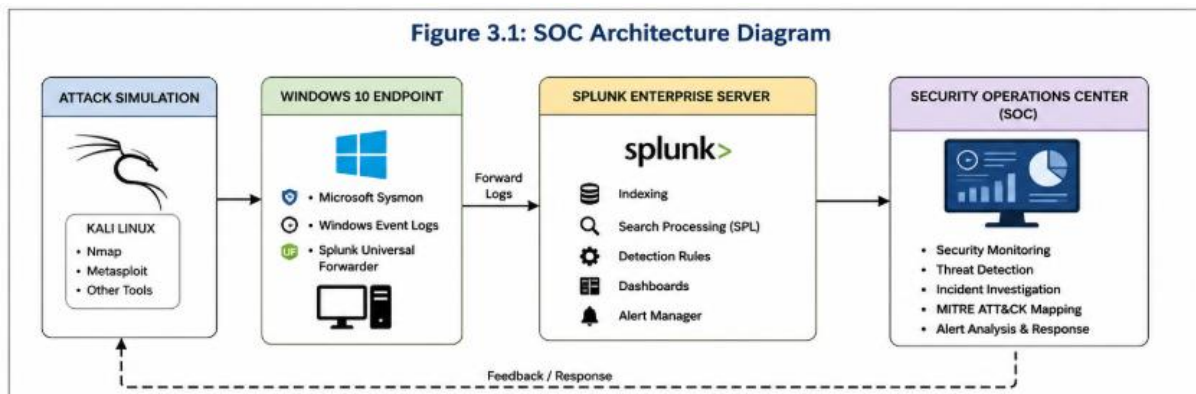
Splunk Universal Forwarder is installed on the Windows machine to securely collect and forward Windows Security Logs and Sysmon Operational Logs to the Splunk Enterprise server.

Splunk Enterprise functions as the centralized Security Information and Event Management (SIEM) platform responsible for:

- Log collection
- Log indexing

- Event correlation
- Threat detection
- Dashboard visualization
- Alert generation

The Kali Linux virtual machine is used to simulate attacker behavior by performing controlled attack scenarios against the Windows endpoint. These simulations validate the effectiveness of the implemented detection rules.



3.3 System Components

3.3.1 Windows 10 Endpoint

The Windows virtual machine serves as the monitored endpoint within the SOC environment. It generates Windows Security Event Logs and Sysmon events based on user activities and simulated attacks.

Installed components include:

- Microsoft Sysmon
- Splunk Universal Forwarder
- Windows Event Logging

The endpoint continuously sends security logs to Splunk Enterprise for analysis.

3.3.2 Splunk Enterprise

Splunk Enterprise serves as the core SIEM platform within the SOC environment.

Its primary responsibilities include:

- Receiving logs from endpoints
- Indexing incoming events

- Executing SPL detection queries
- Creating dashboards
- Triggering alerts
- Supporting incident investigations

Splunk Enterprise provides analysts with centralized visibility into all monitored security events.

3.3.3 Splunk Universal Forwarder

The Splunk Universal Forwarder is a lightweight agent installed on the Windows endpoint.

Its responsibilities include:

- Collecting Windows Event Logs
- Collecting Sysmon logs
- Forwarding logs securely
- Maintaining continuous communication with Splunk Enterprise

The Universal Forwarder minimizes resource consumption while ensuring reliable log transmission.

3.3.4 Microsoft Sysmon

Microsoft Sysmon enhances Windows logging by recording detailed endpoint telemetry.

Important monitored events include:

- Process Creation
- Network Connections
- DNS Queries
- Registry Changes
- File Creation
- Driver Loading

These events provide high-quality telemetry for threat detection.

3.3.5 Kali Linux

Kali Linux is used exclusively for controlled attack simulation.

Several security scenarios were executed, including:

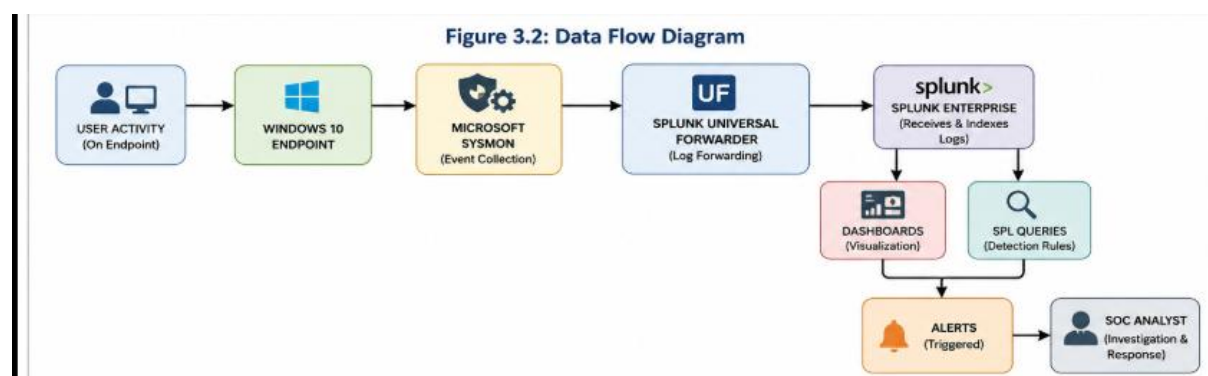
- Failed login attempts
- PowerShell execution
- Network reconnaissance
- Process execution
- DNS activity

These simulations verified the effectiveness of the implemented detection rules.

3.4 Data Flow

The implemented SOC follows the following workflow:

1. User activity occurs on Windows.
2. Windows and Sysmon generate security events.
3. Universal Forwarder collects the logs.
4. Logs are transmitted to Splunk Enterprise.
5. Splunk indexes incoming data.
6. SPL queries analyze security events.
7. Alerts are generated.
8. Dashboards display security information.
9. Security analysts investigate suspicious activities.



3.5 Dashboard Design

Several dashboards were developed within Splunk Enterprise to visualize security events.

The dashboards display:

- Total Events

- Failed Logins
- Successful Logins
- DNS Activity
- Network Connections
- Endpoint Activity
- Process Creation
- Top Hosts
- Top Processes

These visualizations provide analysts with real-time situational awareness.



3.6 Detection Workflow

The detection workflow implemented in this project follows these stages:

- Log Collection
- Event Parsing
- Field Extraction
- SPL Query Execution
- Alert Generation
- Dashboard Visualization
- Incident Investigation

Each detection rule continuously analyzes incoming events to identify suspicious behavior.

3.7 Chapter Summary

This chapter described the architecture and overall design of the SOC Security Monitoring environment. It explained the interaction between Windows, Sysmon, Splunk Universal Forwarder, Splunk Enterprise, and Kali Linux. The next chapter presents the practical implementation of the proposed system.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This chapter describes the practical implementation of the SOC Security Monitoring environment. It covers the installation and configuration of Splunk Enterprise, Microsoft Sysmon, Splunk Universal Forwarder, Windows Event Log collection, dashboard development, detection rule implementation, and alert configuration.

The implementation was performed in a controlled laboratory environment using virtualization technology to simulate real-world SOC operations.

4.2 Laboratory Environment

The project was implemented using the following environment:

Component	Description
Operating System	Windows 10

Component	Description
SIEM Platform	Splunk Enterprise
Log Forwarder	Splunk Universal Forwarder
Endpoint Monitoring	Microsoft Sysmon
Attacker Machine	Kali Linux
Virtualization	Oracle VirtualBox

4.3 Splunk Enterprise Installation

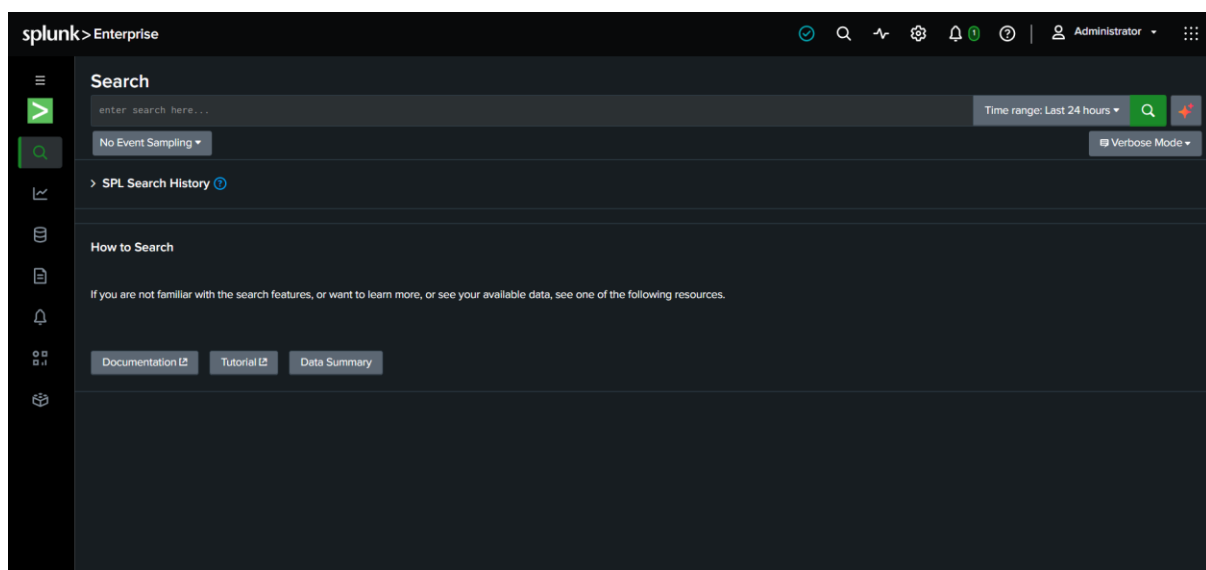
Splunk Enterprise was installed as the centralized SIEM server.

The installation process included:

- Downloading Splunk Enterprise
- Installing the software
- Creating the administrator account
- Configuring web access
- Starting Splunk services
- Verifying successful installation

After installation, the Splunk Web interface was used for data indexing, searching, dashboard creation, and alert configuration.

Insert Figure 4.1: Splunk Enterprise Dashboard



4.4 Windows Endpoint Configuration

The Windows virtual machine was prepared as the monitored endpoint.

Configuration included:

- Windows Updates
- Administrative permissions
- Event Logging
- Network connectivity
- Firewall configuration

The endpoint generated security logs during normal and simulated activities.

4.5 Microsoft Sysmon Installation

Microsoft Sysmon was installed to improve endpoint visibility.

Configuration steps included:

- Downloading Sysmon
- Installing Sysmon
- Applying Sysmon configuration
- Verifying Operational Logs

Sysmon significantly increased the amount of security telemetry available for analysis.

4.6 Splunk Universal Forwarder Configuration

Splunk Universal Forwarder was installed on the Windows endpoint.

Configuration involved:

- Connecting to Splunk Enterprise
- Configuring inputs.conf
- Selecting Windows Event Logs
- Selecting Sysmon Operational Logs
- Starting forwarding services

Successful forwarding was verified using Splunk Search.

4.7 Log Collection

The following logs were successfully collected:

- Windows Security Logs
- Windows System Logs
- Windows Application Logs
- Sysmon Operational Logs

Log validation confirmed successful ingestion into Splunk indexes.

4.8 Dashboard Development

Custom dashboards were created using Splunk Dashboard Studio.

Implemented panels include:

- Security Events
- Login Monitoring
- DNS Activity
- Network Connections
- Endpoint Processes
- Top Hosts
- Top Users
- Alert Overview

These dashboards provide centralized monitoring capabilities.

SOC Dashboard



4.9 Detection Rule Implementation

Six custom SPL detection rules were developed.

Implemented detections include:

- PowerShell Execution
- Failed Login Attempts
- DNS Queries
- Process Creation
- Network Connections
- Suspicious Processes

Each rule continuously monitors incoming events and detects suspicious behavior.

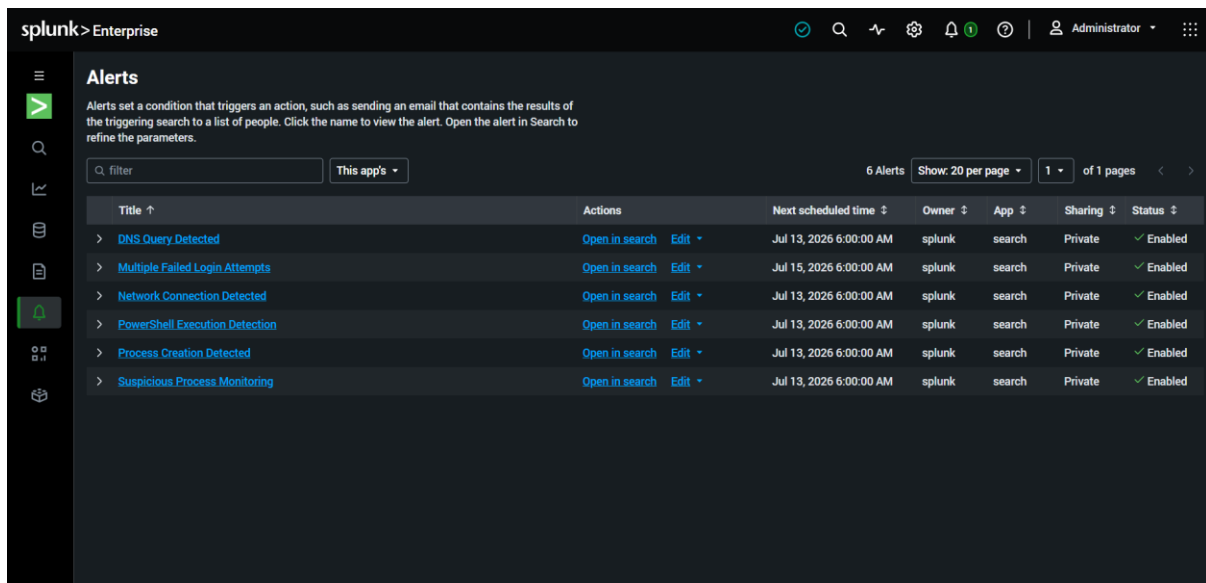
4.10 Alert Configuration

Splunk alerts were configured for each detection rule.

Alert configuration included:

- Scheduled execution
- Trigger conditions
- Severity assignment
- Alert validation

These alerts simulate real-world SOC monitoring.



The screenshot shows the Splunk Alerts interface. At the top, there's a header with 'splunk > Enterprise' and various navigation icons. Below the header, the 'Alerts' section is active, displaying a list of alerts. The alerts are listed in a table with columns: Title, Actions, Next scheduled time, Owner, App, Sharing, and Status. The alerts include 'DNS Query Detected', 'Multiple Failed Login Attempts', 'Network Connection Detected', 'PowerShell Execution Detection', 'Process Creation Detected', and 'Suspicious Process Monitoring'. Each alert has a status of 'Enabled' and a next scheduled time of 'Jul 13, 2026 6:00:00 AM'.

Title ↑	Actions	Next scheduled time ↓	Owner ↓	App ↓	Sharing ↓	Status ↓
> DNS Query Detected	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Multiple Failed Login Attempts	Open in search Edit	Jul 15, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Network Connection Detected	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> PowerShell Execution Detection	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Process Creation Detected	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Suspicious Process Monitoring	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled

4.11 Attack Simulation

Controlled attack scenarios were executed using Kali Linux and Windows.

The simulations included:

- Failed login attempts
- PowerShell execution
- Process execution
- DNS queries
- Network activity

Each simulated activity generated security events that were successfully detected by Splunk.

4.12 Chapter Summary

This chapter presented the practical implementation of the SOC Security Monitoring environment. It described the installation and configuration of Splunk Enterprise, Microsoft Sysmon, Universal Forwarder, log collection, dashboard development, detection rule implementation, and alert configuration. The next chapter focuses on the developed dashboards and their operational capabilities.

CHAPTER 5

DASHBOARD DESIGN AND VISUALIZATION

5.1 Introduction

An effective Security Operations Center (SOC) relies heavily on visualization to monitor security events and support rapid incident response. Dashboards provide security analysts with a centralized view of system activities, enabling them to identify suspicious behavior without manually reviewing thousands of log entries. In this project, custom dashboards were developed using Splunk Enterprise to visualize endpoint telemetry collected from Windows Security Event Logs and Microsoft Sysmon.

The dashboards provide real-time monitoring capabilities by displaying graphical representations of security events, authentication activities, process executions, network connections, and endpoint behavior. These visualizations significantly improve situational awareness and reduce the time required to investigate security incidents.

5.2 Dashboard Objectives

The dashboards developed in this project were designed to achieve the following objectives:

- Provide centralized monitoring of Windows endpoint activities.
- Visualize security events in real time.
- Display authentication activities.
- Monitor PowerShell execution.
- Display process creation events.
- Monitor DNS queries.
- Monitor network connections.
- Provide statistical summaries for SOC analysts.
- Improve incident investigation efficiency.

5.3 Dashboard Components

The implemented SOC dashboard consists of multiple panels that visualize different categories of security events.

5.3.1 Total Security Events

This visualization displays the total number of indexed security events collected by Splunk Enterprise. It provides analysts with an overview of overall system activity and helps identify abnormal increases in event generation.

5.3.2 Authentication Monitoring

Authentication monitoring displays successful and failed login attempts generated from Windows Security Event Logs. Repeated authentication failures may indicate brute-force attacks or unauthorized access attempts.

Displayed information includes:

- Successful Logins
- Failed Logins
- Authentication Trends

5.3.3 Process Monitoring

The dashboard continuously monitors process creation events collected through Microsoft Sysmon.

Information displayed includes:

- Executed Processes
- Parent Processes
- Process Frequency
- Executing User

This panel assists analysts in identifying suspicious applications executed on monitored endpoints.

5.3.4 DNS Activity Monitoring

DNS monitoring visualizes DNS queries generated by endpoint processes.

The dashboard displays:

- Most Requested Domains
- Query Frequency
- Associated Processes

Suspicious DNS requests may indicate malware communication with external command-and-control servers.

_time ↕	host ↕	_raw ↕
2026-07-08T14:31:02.436+05:30	DESKTOP-324D2VJ	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'> <System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}'/> <EventID>22</EventID> <Version>5</Version> <Level>4</Level> <Task>22</Task> <Opcode>0</Opcode> <Keywords>0:8000000000000000</pre>

5.3.5 Network Connection Monitoring

This dashboard visualizes outbound network connections established by endpoint processes.

Displayed information includes:

- Destination IP Address
- Destination Port
- Initiating Process
- Connection Frequency

This assists analysts in identifying abnormal network communication.

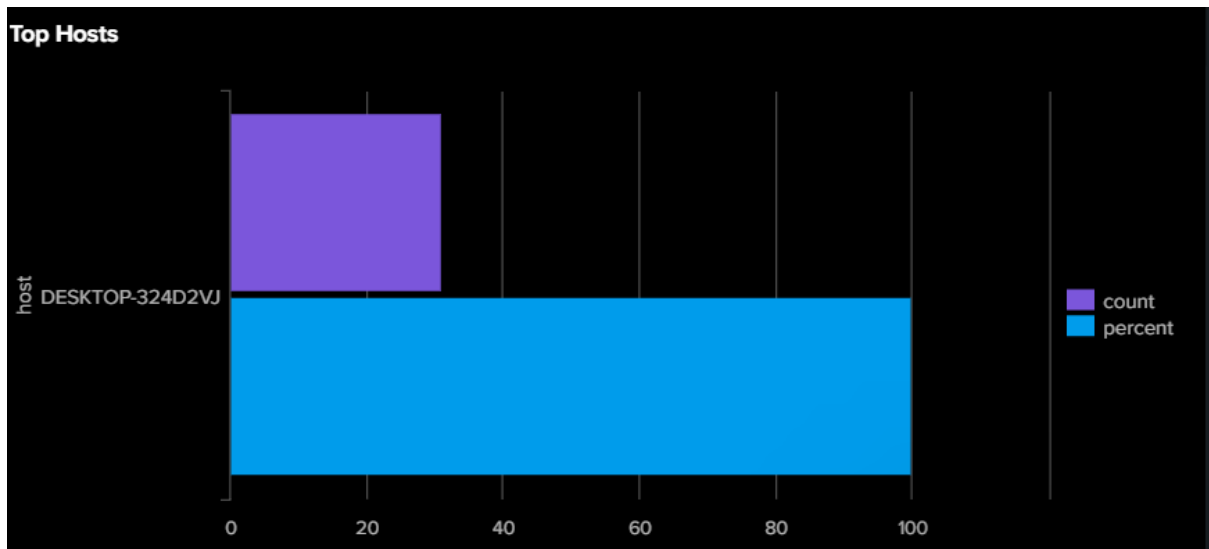
_time ↕	host ↕	_raw ↕
2026-07-08T11:20:42.873+05:30	DESKTOP-324D2VJ	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'> <System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}'/> <EventID>3</EventID> <Version>5</Version> <Level>4</Level><Task>3</Task></pre>

< Prev 1 2 3 4 5 ... Next >

5.3.6 Top Hosts

This visualization identifies systems generating the highest number of security events.

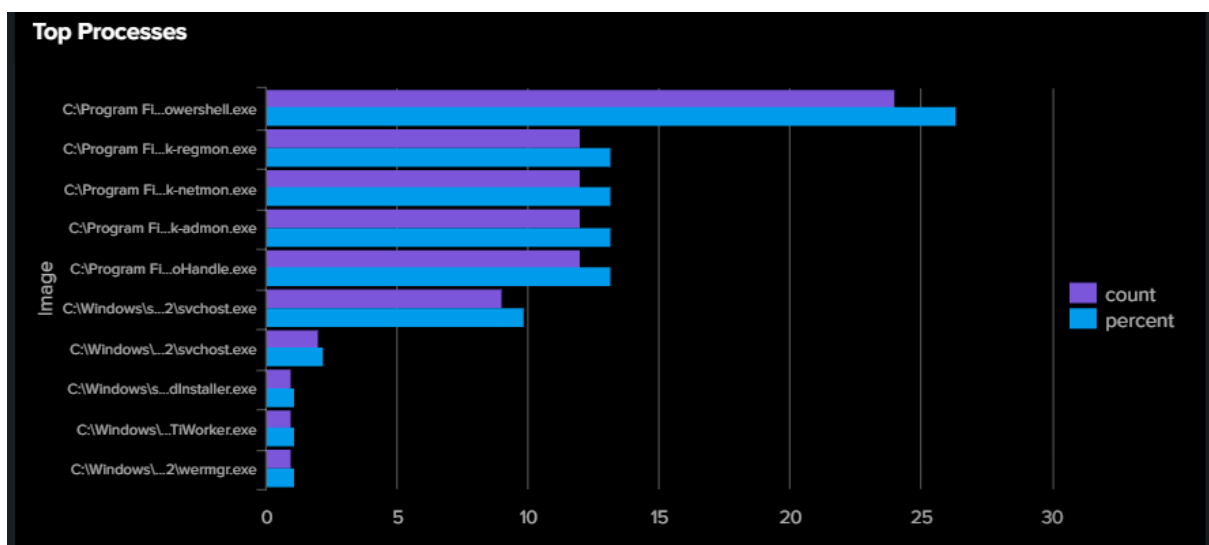
The panel enables analysts to prioritize investigation efforts toward endpoints generating unusually high activity.



5.3.7 Top Processes

This panel displays the most frequently executed processes within the monitored environment.

Frequent execution of administrative utilities such as PowerShell or Command Prompt may require further investigation depending on organizational policies.



5.4 Dashboard Development Process

Dashboard development followed the following stages:

1. Data Collection
2. Field Extraction
3. SPL Query Development

4. Visualization Design
5. Dashboard Integration
6. Validation
7. Performance Optimization

Each visualization was tested using simulated attack scenarios to ensure accurate representation of collected security events.

5.5 Dashboard Benefits

The developed dashboards provide several advantages:

- Centralized monitoring
- Real-time visualization
- Reduced investigation time
- Improved situational awareness
- Simplified incident response
- Enhanced threat detection
- Better analyst productivity

5.6 Dashboard Limitations

Although the dashboards successfully visualize security events, several limitations remain:

- Limited attack simulations
- Single endpoint monitoring
- No external threat intelligence integration
- Small laboratory environment

Future work may integrate additional endpoints, firewall logs, cloud logs, and threat intelligence feeds.

5.7 Chapter Summary

This chapter presented the dashboards developed within Splunk Enterprise. The implemented visualizations provide centralized monitoring of Windows endpoint activities and significantly improve the efficiency of SOC analysts during threat detection and incident investigation.

CHAPTER 6

DETECTION RULES AND SPL QUERIES

6.1 Introduction

Detection rules are the core component of any Security Operations Center. Their primary purpose is to continuously analyze incoming security events and identify suspicious behavior that may indicate cyberattacks or unauthorized activities.

In this project, multiple custom Search Processing Language (SPL) queries were developed within Splunk Enterprise to monitor Windows Security Events and Microsoft Sysmon logs. Each detection rule was designed to identify specific attacker behaviors while minimizing false positives.

6.2 Detection Rule Development Methodology

The detection development process consisted of the following phases:

- Identifying security use cases
- Selecting relevant Windows Event IDs
- Selecting relevant Sysmon Event IDs
- Developing SPL queries
- Testing detections
- Mapping detections to MITRE ATT&CK
- Configuring alerts
- Validating detection accuracy

Each detection rule was validated using simulated endpoint activities.

6.3 PowerShell Execution Detection

Objective

Detect execution of PowerShell on monitored Windows endpoints.

Data Source

Microsoft Sysmon Event ID 1

SPL Query

```
index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
```

| search powershell

| table _time host User Image CommandLine

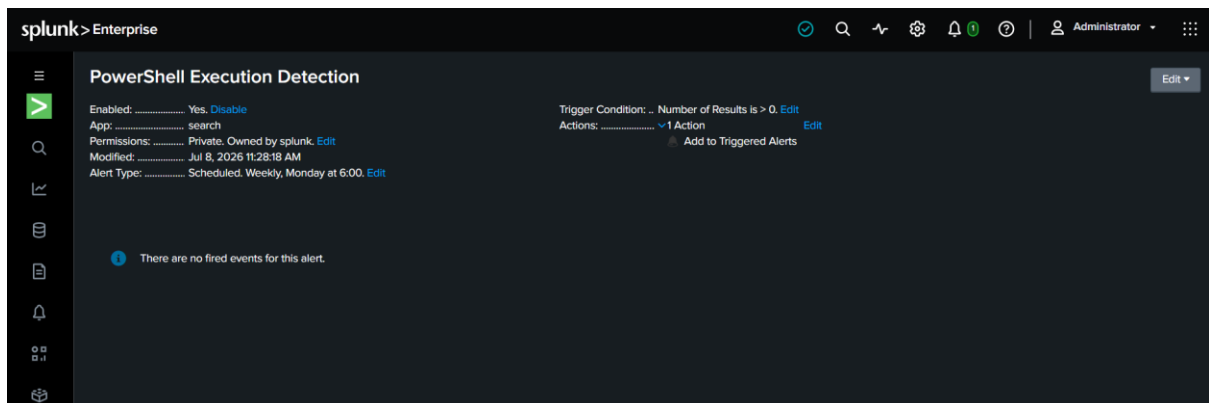
Detection Logic

The rule monitors process creation events and identifies PowerShell execution.

MITRE ATT&CK

Execution

T1059.001 – PowerShell



6.4 Failed Login Detection

Objective

Detect repeated Windows authentication failures.

Data Source

Windows Security Event ID 4625

SPL Query

index=* EventCode=4625

| stats count by Account_Name

Detection Logic

Monitors failed authentication attempts to identify brute-force attacks.

MITRE ATT&CK

Credential Access

T1110 – Brute Force

splunk> Enterprise

PowerShell Execution Detection

Save Save As View Create Table View Close

Time range: Last 1 week

```

index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
| rex field=_raw "<Data Name='Image'>{?<Image>.*}</Data>"
| rex field=_raw "<Data Name='CommandLine'>{?<CommandLine>.*}</Data>"
| rex field=_raw "<Data Name='User'>{?<User>.*}</Data>"
| search Image="powershell.exe"
| search NOT Image="*splunk-powershell.exe"
| table _time host User Image CommandLine

```

12 events (7/2/26 2:54:45.000 PM to 7/9/26 2:54:45.000 PM) No Event Sampling

Events (12) Patterns Statistics (12) Visualization

Show: 20 Per Page Format Preview: On

_time	host	User	Image	CommandLine
2026-07-08 12:58:59.313	DESKTOP-32402VJ	DESKTOP-32402VJ\barkh	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-08 12:58:57.067	DESKTOP-32402VJ	DESKTOP-32402VJ\barkh	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2026-07-08 11:22:23.879	DESKTOP-32402VJ	DESKTOP-32402VJ\barkh	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-08 11:22:18.193	DESKTOP-32402VJ	DESKTOP-32402VJ\barkh	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2026-07-08 11:28:54.997	DESKTOP-32402VJ	NT AUTHORITY\SYSTEM	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-08 11:28:53.206	DESKTOP-32402VJ	NT AUTHORITY\SYSTEM	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe -ExecutionPolicy Restricted -Command Write-Host 'Final result: 1!';

6.5 DNS Query Detection

Objective

Monitor suspicious DNS requests generated by endpoint processes.

Data Source

Sysmon Event ID 22

SPL Query

index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"

EventID=22

| stats count by QueryName Image

MITRE ATT&CK

Command and Control

T1071.004 – DNS

splunk> Enterprise

DNS Query Detected Save Save As View Create Table View Close

index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
 | rex field=_raw "<EventID>(<?EventID>\d+)</EventID>"
 | search EventID=22 Time range: Last 1 week

94 events (7/2/26 2:55:20.000 PM to 7/9/26 2:55:20.000 PM) No Event Sampling Job

Events (94) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 50 Per Page View: List

Time	Event
7/8/26 2:31:02.436 PM	<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Sysmon" Guid="{5770385f-c27a-43e0-bf4c-86f5698fbd9}" /><EventID>22</EventID><Version>5</Version><Level>4</Level><Task>22</Task><Opcode>0</Opcode><Keywords>0x80000000</Keywords><TimeCreated SystemTime="2026-07-08T09:01:02.4362484Z" /><EventRecordID>5270</EventRecordID><Correlation><Execution ProcessID="2192" ThreadID="3744" /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-324D2VJ</Computer><Security UserID="S-1-5-18" /></System><EventData><Data Name="RuleName"></Data><Data Name="UtcTime">2026-07-08 07:46:12.984</Data><Data Name="Process Guid">{5f41928b-e4be-6a4d-c900-000000000000}</Data><Data Name="ProcessId">1772</Data><Data Name="QueryName">my.microsoftpersonalcontent.c om</Data><Data Name="QueryStatus">1460</Data><Data Name="QueryResults"></Data><Data Name="Image">C:\Users\barkh\AppData\Local\Microsoft \OneDrive\OneDrive.exe</Data><Data Name="User">DESKTOP-324D2VJ\barkh</Data></EventData></Event> host = DESKTOP-324D2VJ source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
7/8/26 12:37:41.511 PM	<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Sysmon" Guid="{5770385f-c27a-43e0-bf4c-86f5698fbd9}" /><EventID>22</EventID><Version>5</Version><Level>4</Level><Task>22</Task><Opcode>0</Opcode><Keywords>0x80000000</Keywords><TimeCreated SystemTime="2026-07-08T07:07:41.5114690Z" /><EventRecordID>4979</EventRecordID><Correlation><Execution

6.6 Process Creation Detection

Objective

Monitor newly created processes.

Data Source

Sysmon Event ID 1

SPL Query

index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"

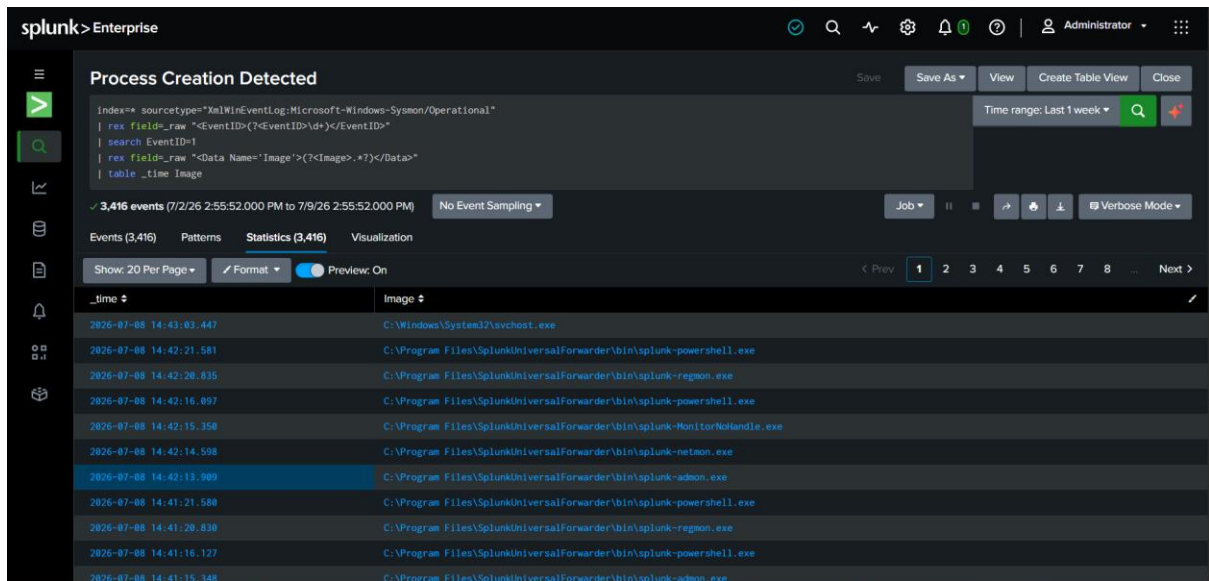
EventID=1

| stats count by Image User

MITRE ATT&CK

Execution

T1204 – User Execution



6.7 Network Connection Detection

Objective

Detect outbound network connections.

Data Source

Sysmon Event ID 3

SPL Query

`index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"`

`EventID=3`

`| stats count by Image DestinationIp DestinationPort`

MITRE ATT&CK

Command and Control

T1071 – Application Layer Protocol

6.8 Suspicious Process Detection

Objective

Detect execution of commonly abused administrative tools.

Monitored Processes

- PowerShell
- CMD
- WMIC

- PsExec

SPL Query

index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"

EventID=1

(Image="*cmd.exe" OR Image="*powershell.exe" OR Image="*wmic.exe")

| table _time host User Image CommandLine

MITRE ATT&CK

Execution

T1059 – Command and Scripting Interpreter

6.9 Detection Rule Summary

Detection Rule	Data Source	MITRE ATT&CK
PowerShell Execution	Sysmon Event ID 1	T1059.001
Failed Login	Windows Event ID 4625	T1110
DNS Query	Sysmon Event ID 22	T1071.004
Process Creation	Sysmon Event ID 1	T1204
Network Connection	Sysmon Event ID 3	T1071
Suspicious Process	Sysmon Event ID 1	T1059

6.10 Chapter Summary

This chapter presented the custom SPL detection rules developed for the SOC environment. Each rule was designed to monitor a specific attack technique and was mapped to the corresponding MITRE ATT&CK technique. These detections form the analytical core of the SOC implementation and enable real-time identification of suspicious endpoint activities.

CHAPTER 7

ALERT CONFIGURATION AND MITRE ATT&CK MAPPING

7.1 Introduction

Detection rules alone are insufficient for effective Security Operations Center (SOC) monitoring. To enable real-time incident detection and rapid response, security events must automatically generate alerts whenever suspicious activities are identified. Splunk Enterprise provides flexible alerting capabilities that allow SOC analysts to receive notifications whenever predefined detection conditions are met.

In addition, mapping detection rules to the MITRE ATT&CK Framework provides standardized classification of attacker behavior, helping analysts understand adversary tactics, techniques, and procedures (TTPs). This chapter discusses the implementation of Splunk alerts and the mapping of detection rules to MITRE ATT&CK techniques.

7.2 Alert Configuration in Splunk Enterprise

Splunk Enterprise provides scheduled and real-time alerting mechanisms. In this project, each custom detection rule was configured as an individual alert to continuously monitor incoming security events.

Each alert consists of:

- Alert Name
- SPL Search
- Trigger Condition
- Alert Frequency
- Severity
- Notification Action

The alerts were configured using the "Save As Alert" feature available in Splunk Enterprise.

7.3 Configured Alerts

The following alerts were successfully implemented.

Alert Name	Purpose
PowerShell Execution Detection	Detect PowerShell execution
Failed Login Detection	Detect brute-force attempts
DNS Query Detection	Monitor DNS activity
Process Creation Detection	Monitor new processes
Network Connection Detection	Detect outbound connections

Alert Name	Purpose
Suspicious Process Detection	Detect execution of suspicious utilities

7.4 Alert Workflow

The implemented alert workflow follows the sequence below:

1. Windows generates security events.
2. Sysmon records endpoint telemetry.
3. Universal Forwarder sends logs to Splunk.
4. Splunk indexes incoming events.
5. SPL queries analyze the indexed data.
6. Detection rule conditions are evaluated.
7. Alerts are generated.
8. Analysts investigate suspicious activities.

This workflow enables automated security monitoring with minimal manual intervention.

7.5 MITRE ATT&CK Framework

The MITRE ATT&CK Framework is widely adopted for classifying attacker behavior. It organizes cyberattack techniques according to the attack lifecycle and assists SOC analysts in understanding adversary activities.

Mapping detection rules to MITRE ATT&CK provides several advantages:

- Standardized threat classification
- Improved detection coverage
- Better incident reporting
- Easier threat hunting
- Enhanced SOC maturity

7.6 MITRE ATT&CK Mapping

The developed detection rules were mapped to the following MITRE ATT&CK techniques.

MITRE ATT&CK Mapping of Detection Rules

Detection Rule	SPL/Data Source	MITRE ATT&CK	Technique ID	Tactic
PowerShell Execution Detection	Sysmon Event ID 1	PowerShell	T1059.001	Execution
Multiple Failed Login Attempts	Windows Event ID 4625	Brute Force	T1110	Credential Access
Process Creation Detection	Sysmon Event ID 1	Process Discovery	T1057	Discovery
Network Connection Detection	Sysmon Event ID 3	System Network Connections Discovery	T1049	Discovery
DNS Query Detection	Sysmon Event ID 22	DNS	T1071.004	Command & Control
Suspicious Process Monitoring	Sysmon Event ID 1	Masquerading	T1036	Defense Evasion
Registry Monitoring	Sysmon Event ID 13	Modify Registry	T1112	Defense Evasion
Registry Object Creation	Sysmon Event ID 12	Modify Registry	T1112	Defense Evasion
Driver/Configuration Change	Sysmon Event ID 16	Modify Registry	T1112	Defense Evasion

7.7 Benefits of Alerting

The configured alerts provide several operational benefits:

- Continuous monitoring
- Faster incident detection
- Reduced analyst workload
- Automated threat identification
- Improved security visibility
- Centralized event management

7.8 Chapter Summary

This chapter discussed the implementation of automated alerting within Splunk Enterprise and demonstrated how each custom detection rule was mapped to the MITRE ATT&CK Framework. These features significantly enhance the analytical capabilities of the SOC by enabling rapid detection and standardized threat classification.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Introduction

The objective of this project was to implement a functional Security Operations Center (SOC) using Splunk Enterprise capable of collecting security logs, detecting suspicious

activities, generating alerts, and providing centralized monitoring. This chapter presents the results obtained during implementation and evaluates the effectiveness of the developed detection rules and dashboards.

8.2 Successful Deployment

The SOC laboratory was successfully deployed using the following components:

- Splunk Enterprise
- Windows 10
- Splunk Universal Forwarder
- Microsoft Sysmon
- Kali Linux

All components communicated successfully, enabling centralized log collection and analysis.

8.3 Log Collection Results

Splunk Enterprise successfully received:

- Windows Security Event Logs
- Windows System Logs
- Windows Application Logs
- Microsoft Sysmon Operational Logs

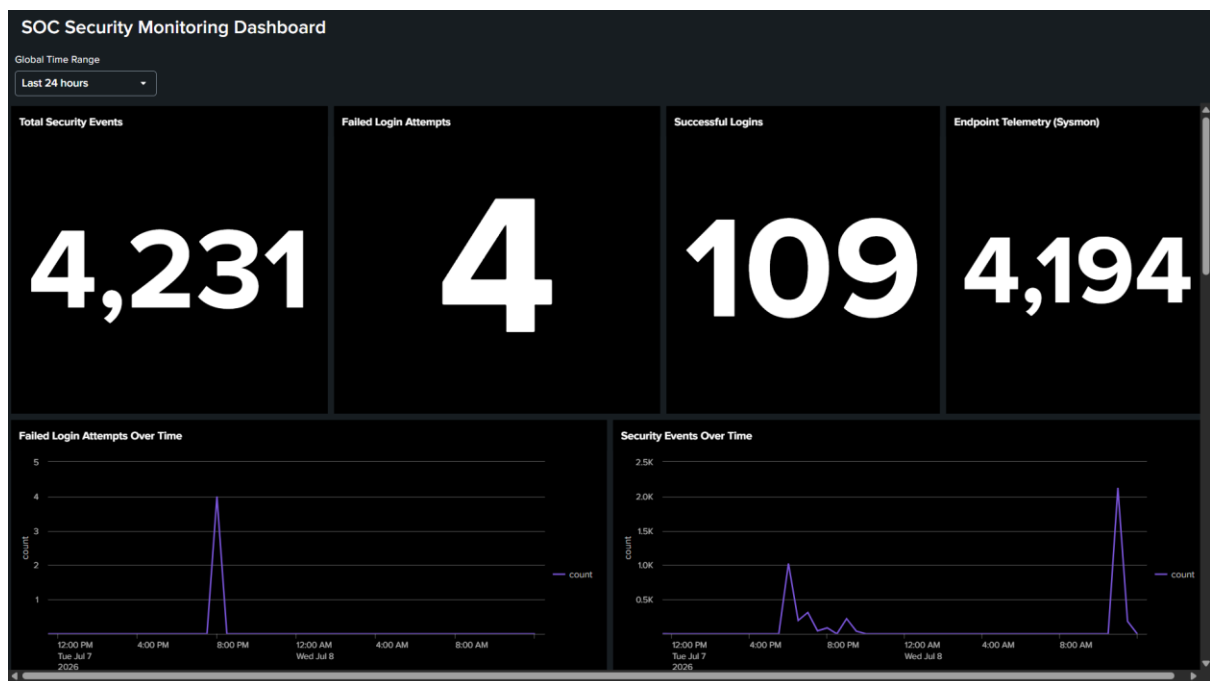
Continuous log forwarding confirmed successful integration between the Windows endpoint and Splunk Enterprise.

8.4 Dashboard Results

The dashboards successfully displayed:

- Total Security Events
- Authentication Activity
- DNS Queries
- Network Connections
- Endpoint Processes
- Top Hosts
- Top Users

The visualizations significantly simplified monitoring and event analysis.



8.5 Detection Results

Each custom SPL query successfully detected the intended activities.

Successfully detected events included:

- PowerShell execution
- Failed login attempts
- DNS queries
- Process creation
- Network connections
- Suspicious process execution

Detection accuracy was verified using simulated endpoint activities.

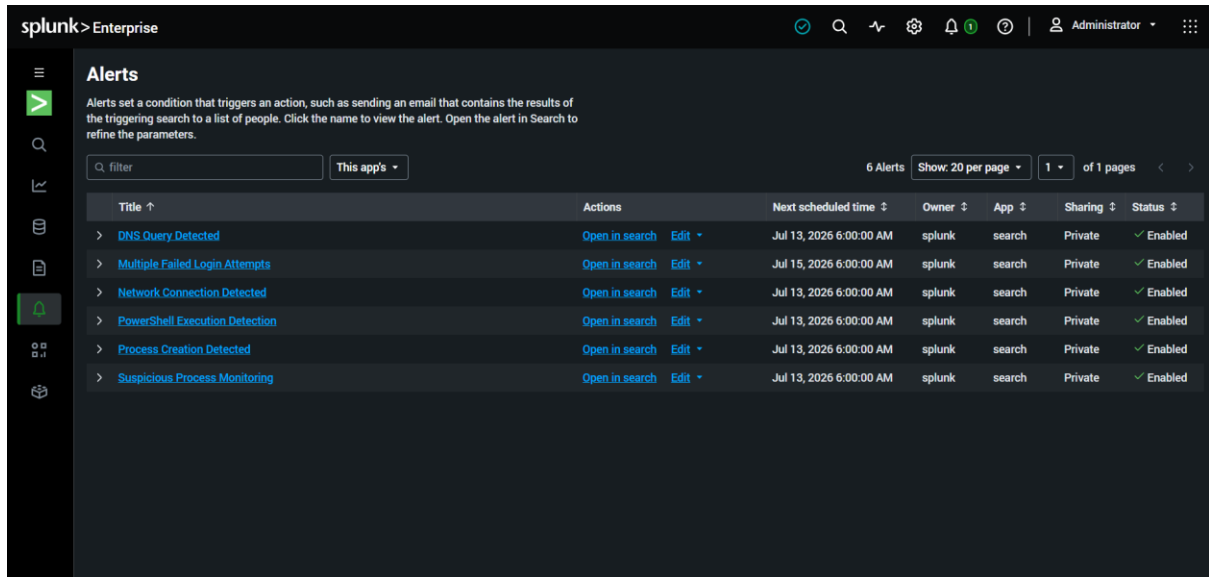
8.6 Alert Results

Every configured alert successfully triggered whenever its associated detection rule matched incoming events.

Generated alerts included:

- PowerShell Alert
- Failed Login Alert
- DNS Query Alert

- Process Creation Alert
- Network Alert
- Suspicious Process Alert



The screenshot shows the Splunk Alerts interface. At the top, there's a header with 'splunk > Enterprise' and various navigation icons. Below the header, a sidebar on the left contains icons for different Splunk views. The main area is titled 'Alerts' and includes a brief description: 'Alerts set a condition that triggers an action, such as sending an email that contains the results of the triggering search to a list of people. Click the name to view the alert. Open the alert in Search to refine the parameters.' Below this is a search filter and a 'This app's' dropdown. A table lists 6 alerts, showing details like title, actions, next scheduled time, owner, app, sharing, and status.

Title ↑	Actions	Next scheduled time ↑	Owner ↑	App ↑	Sharing ↑	Status ↑
> DNS Query Detected	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Multiple Failed Login Attempts	Open in search Edit	Jul 15, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Network Connection Detected	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> PowerShell Execution Detection	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Process Creation Detected	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled
> Suspicious Process Monitoring	Open in search Edit	Jul 13, 2026 6:00:00 AM	splunk	search	Private	✓ Enabled

8.7 Discussion

The implemented SOC successfully demonstrated the core responsibilities of a modern Security Operations Center.

Key achievements include:

- Centralized security monitoring
- Endpoint visibility
- Automated detection
- Dashboard visualization
- Alert generation
- Threat investigation

The project successfully illustrates how SIEM technology improves cybersecurity monitoring through automation and centralized analysis.

8.8 Limitations

Although the project achieved its objectives, several limitations remain:

- Single Windows endpoint
- Small laboratory environment
- Limited attack simulations

- No email notification integration
- No external threat intelligence feeds

Future implementations may integrate cloud environments, multiple endpoints, and advanced threat intelligence platforms.

8.9 Chapter Summary

The obtained results demonstrate that Splunk Enterprise can effectively function as a centralized SIEM platform for SOC monitoring. The developed dashboards, detection rules, and alerts successfully detected simulated security activities while providing valuable visibility into endpoint behavior.

CHAPTER 9

CONCLUSION AND FUTURE WORK

9.1 Conclusion

This project successfully implemented a Security Operations Center (SOC) laboratory using Splunk Enterprise, Microsoft Sysmon, Splunk Universal Forwarder, and Windows Event Logs. The implemented environment demonstrated centralized log collection, security monitoring, threat detection, dashboard visualization, and automated alert generation.

Throughout the project, Windows Security Events and Sysmon telemetry were collected and analyzed using custom Search Processing Language (SPL) queries. Multiple detection rules were developed to identify suspicious activities, including PowerShell execution, failed login attempts, DNS queries, process creation, network connections, and suspicious process execution. These detections were successfully mapped to the MITRE ATT&CK Framework to improve threat classification and align the monitoring environment with industry best practices.

The developed dashboards provided centralized visualization of security events, allowing analysts to monitor endpoint activity efficiently. Automated alerts further enhanced the effectiveness of the SOC by enabling rapid identification of suspicious behavior.

Overall, the project achieved all proposed objectives and provided practical experience in SIEM deployment, log analysis, threat detection, dashboard development, and incident investigation. The implemented laboratory closely simulates real-world SOC operations and provides a strong foundation for advanced cybersecurity research and professional practice.

9.2 Future Work

The functionality of the implemented SOC can be further enhanced through several improvements.

Future enhancements include:

- Monitoring multiple Windows endpoints
- Integrating Linux log collection
- Deploying email notifications for alerts
- Implementing threat intelligence feeds
- Integrating Sigma detection rules
- Monitoring cloud environments
- Applying machine learning for anomaly detection
- Implementing automated incident response
- Integrating SOAR platforms
- Expanding dashboard analytics

These enhancements would significantly improve detection capabilities while bringing the laboratory closer to enterprise SOC environments.

9.3 Overall Project Contribution

The project demonstrates practical implementation of several cybersecurity concepts, including:

- Security Information and Event Management (SIEM)
- Security Operations Center (SOC)
- Endpoint Monitoring
- Threat Detection
- Log Analysis
- Incident Investigation
- MITRE ATT&CK Mapping
- Dashboard Development
- Alert Configuration
- Security Monitoring

The knowledge and practical experience gained through this implementation contribute significantly to the understanding of modern cybersecurity operations and enterprise security monitoring.

9.4 Final Remarks

The increasing sophistication of cyber threats requires organizations to adopt proactive monitoring solutions capable of detecting malicious activities in real time. SIEM platforms such as Splunk Enterprise play a critical role in modern Security Operations Centers by providing centralized visibility, advanced analytics, and automated alerting.

This project demonstrates how enterprise-grade monitoring can be implemented within a laboratory environment while following industry best practices. The developed SOC laboratory serves as an effective educational platform for learning threat detection, security monitoring, and incident response.

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